

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Clarification for DFP State Machine for Adaptive Uni-Directional Time Sync Handshake
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

This ECN adds clarification for the DFP State Machine for Adaptive Uni-Directional Time Sync Handshake description in the spec without any functional behavior change.
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Benefits as a result of the changes:

Clarifies the spec description.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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NA

An analysis of the hardware implications:
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NA

An analysis of the software implications:
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NA

An analysis of the compliance testing implications:
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NA

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Actual Change

(a). Section 7.3.1.3.2 Adaptive Uni-Directional Mode

To Text:

When an Adapter is in the CL2 state, a Downstream Facing Port shall stop Time Sync Handshakes. When ~~exiting~~exited CL2 state back to CL0 state, the Time Sync Handshakes shall start in Inversed Bi-Directional mode as if the TMU was enabled, including objecting to CLx states before the transition to Adaptive Uni-Directional mode. A Router shall reach the required time synchronization accuracy within tConvergeTime after Time Sync Handshakes are resumed when exiting CL2.

Note: During the transition to CL0, DFP and UFP may not synchronize perfectly. Consequently, DFP handshakes initiated immediately after entering CL0 could time out if UFP is still transitioning from CL2 to CL0.

Figure 7-139 shows the state machine that is recommended for an Upstream Facing Port using Adaptive Uni-Directional Time Sync Handshakes. The state machine uses the following timeout values: